Assignment_2_ML

June 23, 2023

[ ]:
```
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```

[1]:
```
#Q1

salary = float(input("Enter your salary: "))
years_of_service = int(input("Enter your years of service: "))

if years_of_service > 5:
    bonus_percentage = 10
else:
    bonus_percentage = 5

net_bonus_amount = salary * (bonus_percentage / 100)

print("Net bonus amount: Rs." + str(net_bonus_amount))
```

```
Enter your salary:  1084174104
Enter your years of service:  13214

Net bonus amount: Rs.108417410.4
```

[2]:
```
#Q2

length = float(input("Enter the length of the rectangle: "))
breadth = float(input("Enter the breadth of the rectangle: "))

if length == breadth:
    print("The rectangle is a square.")
else:
    print("The rectangle is not a square.")
```

```
Enter the length of the rectangle:  33
Enter the breadth of the rectangle:  33

The rectangle is a square.
```

[4]:
```
#Q3
```

```python
marks = float(input("Enter your marks: "))

if marks < 25:
    grade = "F"
elif marks < 45:
    grade = "E"
elif marks < 50:
    grade = "D"
elif marks < 60:
    grade = "C"
elif marks < 80:
    grade = "B"
else:
    grade = "A"

print("Your grade is:", grade)
```

```
Enter your marks:  22
```

```
Your grade is: F
```

[5]:

```python
#Q4.1

numbers = list(range(1, 101))

for number in numbers:
    if number % 5 == 0:
        print(number)
```

```
5
10
15
20
25
30
35
40
45
50
55
60
65
70
75
80
85
90
95
```

100

[6]:
```
#Q4.2

numbers = list(range(1, 101))
divisible_by_5 = [number for number in numbers if number % 5 == 0]

print(divisible_by_5)
```

```
[5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100]
```

[7]:
```
#Q4.3

def get_numbers_divisible_by_5():
    numbers = list(range(1, 101))
    divisible_by_5 = [number for number in numbers if number % 5 == 0]
    return divisible_by_5

result = get_numbers_divisible_by_5()
print(result)
```

```
[5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100]
```

[8]:
```
#Q5

odd_numbers = [num for num in range(500, 1001) if num % 2 != 0]
even_numbers = [num for num in range(500, 1001) if num % 2 == 0]

print("Odd numbers:", odd_numbers)
print("Even numbers:", even_numbers)
```

```
Odd numbers: [501, 503, 505, 507, 509, 511, 513, 515, 517, 519, 521, 523, 525,
527, 529, 531, 533, 535, 537, 539, 541, 543, 545, 547, 549, 551, 553, 555, 557,
559, 561, 563, 565, 567, 569, 571, 573, 575, 577, 579, 581, 583, 585, 587, 589,
591, 593, 595, 597, 599, 601, 603, 605, 607, 609, 611, 613, 615, 617, 619, 621,
623, 625, 627, 629, 631, 633, 635, 637, 639, 641, 643, 645, 647, 649, 651, 653,
655, 657, 659, 661, 663, 665, 667, 669, 671, 673, 675, 677, 679, 681, 683, 685,
687, 689, 691, 693, 695, 697, 699, 701, 703, 705, 707, 709, 711, 713, 715, 717,
719, 721, 723, 725, 727, 729, 731, 733, 735, 737, 739, 741, 743, 745, 747, 749,
751, 753, 755, 757, 759, 761, 763, 765, 767, 769, 771, 773, 775, 777, 779, 781,
783, 785, 787, 789, 791, 793, 795, 797, 799, 801, 803, 805, 807, 809, 811, 813,
815, 817, 819, 821, 823, 825, 827, 829, 831, 833, 835, 837, 839, 841, 843, 845,
847, 849, 851, 853, 855, 857, 859, 861, 863, 865, 867, 869, 871, 873, 875, 877,
879, 881, 883, 885, 887, 889, 891, 893, 895, 897, 899, 901, 903, 905, 907, 909,
911, 913, 915, 917, 919, 921, 923, 925, 927, 929, 931, 933, 935, 937, 939, 941,
943, 945, 947, 949, 951, 953, 955, 957, 959, 961, 963, 965, 967, 969, 971, 973,
975, 977, 979, 981, 983, 985, 987, 989, 991, 993, 995, 997, 999]
Even numbers: [500, 502, 504, 506, 508, 510, 512, 514, 516, 518, 520, 522, 524,
```

```
526, 528, 530, 532, 534, 536, 538, 540, 542, 544, 546, 548, 550, 552, 554, 556,
558, 560, 562, 564, 566, 568, 570, 572, 574, 576, 578, 580, 582, 584, 586, 588,
590, 592, 594, 596, 598, 600, 602, 604, 606, 608, 610, 612, 614, 616, 618, 620,
622, 624, 626, 628, 630, 632, 634, 636, 638, 640, 642, 644, 646, 648, 650, 652,
654, 656, 658, 660, 662, 664, 666, 668, 670, 672, 674, 676, 678, 680, 682, 684,
686, 688, 690, 692, 694, 696, 698, 700, 702, 704, 706, 708, 710, 712, 714, 716,
718, 720, 722, 724, 726, 728, 730, 732, 734, 736, 738, 740, 742, 744, 746, 748,
750, 752, 754, 756, 758, 760, 762, 764, 766, 768, 770, 772, 774, 776, 778, 780,
782, 784, 786, 788, 790, 792, 794, 796, 798, 800, 802, 804, 806, 808, 810, 812,
814, 816, 818, 820, 822, 824, 826, 828, 830, 832, 834, 836, 838, 840, 842, 844,
846, 848, 850, 852, 854, 856, 858, 860, 862, 864, 866, 868, 870, 872, 874, 876,
878, 880, 882, 884, 886, 888, 890, 892, 894, 896, 898, 900, 902, 904, 906, 908,
910, 912, 914, 916, 918, 920, 922, 924, 926, 928, 930, 932, 934, 936, 938, 940,
942, 944, 946, 948, 950, 952, 954, 956, 958, 960, 962, 964, 966, 968, 970, 972,
974, 976, 978, 980, 982, 984, 986, 988, 990, 992, 994, 996, 998, 1000]
```

[16]:

```python
#Q6

def count_occurrences(elements, target):
    count = 0
    for element in elements:
        if element == target:
            count += 1
    return count

my_list = input("Enter elements separated by spaces: ").split()
my_list = list(my_list)

print("The list:", my_list)

target_element = input("Enter the Target element: ")

result = count_occurrences(my_list, target_element)
print("The target element appears", result, "times in the list.")
```

```
Enter elements separated by spaces:  11 22 33 44 55 55 55

The list: ['11', '22', '33', '44', '55', '55', '55']

Enter the Target element:  55

The target element appears 3 times in the list.
```

[18]:

```python
#Q7

def sort_words(paragraph):
    words = paragraph.split()
    sorted_words = sorted(words)
    return sorted(sorted_words)
```

```
input_paragraph = input("Enter the paragraph: ")

result = sort_words(input_paragraph)
print("Sorted words:", result)
```

```
Enter the paragraph:  Batman is a superhero appearing in American comic books
published by DC Comics. The character was created by artist Bob Kane and writer
Bill Finger, and debuted in the 27th issue of the comic book Detective Comics on
March 30, 1939. In the DC Universe continuity, Batman is the alias of Bruce
Wayne, a wealthy American playboy, philanthropist, and industrialist who resides
in Gotham City. Batman's origin story features him swearing vengeance against
criminals after witnessing the murder of his parents Thomas and Martha as a
child, a vendetta tempered with the ideal of justice. He trains himself
physically and intellectually, crafts a bat-inspired persona, and monitors the
Gotham streets at night. Kane, Finger, and other creators accompanied Batman
with supporting characters, including his sidekicks Robin and Batgirl; allies
Alfred Pennyworth, James Gordon, and Catwoman; and foes such as the Penguin, the
Riddler, Two-Face, and his archenemy, the Joker.
```

```
Sorted words: ['1939.', '27th', '30,', 'Alfred', 'American', 'American',
'Batgirl;', 'Batman', 'Batman', 'Batman', "Batman's", 'Bill', 'Bob', 'Bruce',
'Catwoman;', 'City.', 'Comics', 'Comics.', 'DC', 'DC', 'Detective', 'Finger,',
'Finger,', 'Gordon,', 'Gotham', 'Gotham', 'He', 'In', 'James', 'Joker.', 'Kane',
'Kane,', 'March', 'Martha', 'Penguin,', 'Pennyworth,', 'Riddler,', 'Robin',
'The', 'Thomas', 'Two-Face,', 'Universe', 'Wayne,', 'a', 'a', 'a', 'a', 'a',
'accompanied', 'after', 'against', 'alias', 'allies', 'and', 'and', 'and',
'and', 'and', 'and', 'and', 'and', 'and', 'and', 'and', 'appearing',
'archenemy,', 'artist', 'as', 'as', 'at', 'bat-inspired', 'book', 'books', 'by',
'by', 'character', 'characters,', 'child,', 'comic', 'comic', 'continuity,',
'crafts', 'created', 'creators', 'criminals', 'debuted', 'features', 'foes',
'him', 'himself', 'his', 'his', 'his', 'ideal', 'in', 'in', 'in', 'including',
'industrialist', 'intellectually,', 'is', 'is', 'issue', 'justice.', 'monitors',
'murder', 'night.', 'of', 'of', 'of', 'of', 'on', 'origin', 'other', 'parents',
'persona,', 'philanthropist,', 'physically', 'playboy,', 'published', 'resides',
'sidekicks', 'story', 'streets', 'such', 'superhero', 'supporting', 'swearing',
'tempered', 'the', 'the', 'the', 'the', 'the', 'the', 'the', 'the', 'the',
'the', 'trains', 'vendetta', 'vengeance', 'was', 'wealthy', 'who', 'with',
'with', 'witnessing', 'writer']
```

[19]:
```
#Q8

mixed = [1, 2, 'abc', 7.34, 8, 'opq', 9.99, 3.14, 'xyz', 99]

integers = []
floats = []
strings = []
```

```python
for item in mixed:
    if isinstance(item, int):
        integers.append(item)
    elif isinstance(item, float):
        floats.append(item)
    elif isinstance(item, str):
        strings.append(item)

print("Integers:", integers)
print("Floats:", floats)
print("Strings:", strings)
```

```
Integers: [1, 2, 8, 99]
Floats: [7.34, 9.99, 3.14]
Strings: ['abc', 'opq', 'xyz']
```